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Sistema de Dosimetria *Wireless* para Radiologia de Intervenção

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Projeto apresentado à Universidade de Aveiro para cumprimento dos requisitos necessários à aprovação na unidade curricular de Laboratórios Avançados de Engenharia Biomédica, realizada sob a orientação científica do Doutor Carlos Davide da Rocha Azevedo, professor auxiliar do Departamento de Física da Universidade de Aveiro

o júri / the jury

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Palavras-chave

Dosímetros pessoais ativos, Exposição ocupacional, Monitorização em tempo real, Radiação dispersa, Radiologia de intervenção.

Resumo

A radiologia de intervenção (RI) é a especialidade médica sujeita à maior exposição profissional a radiação ionizante, dado depender da fluoroscopia. Nesta modalidade, a radiação dispersa é o principal fator que contribui para a dose do operador. Atualmente, a monitorização dos trabalhadores reside em dosímetros passivos que apenas fornecem informação *a posteriori* da dose acumulada. Como consequência, a capacidade de prevenir e agir sobre exposições desnecessárias é limitada.

Os dosímetros pessoais ativos são considerados uma solução complementar. Estes sistemas fornecem informação em tempo real, possibilitando mudanças comportamentais preventivas. Este trabalho analisa o atual estado da arte dos dosímetros ativos na RI. São apresentados os equipamentos disponíveis no mercado e a sua evolução, bem como os princípios de funcionamento. Benefícios documentados na literatura incluem uma redução geral da dose, dependente das condições base de exposição, e maior consciencialização da equipa quanto à radiação recebida durante as intervenções. O seu papel como ferramenta educativa para boas práticas de proteção radiológica também é reconhecido. Contudo, os dispositivos existentes têm um elevado custo associado e são limitados à monitorização ao nível do tórax. Como resposta, é proposto o desenvolvimento de um dosimetro ativo durante o próximo ano letivo. O sistema é projetado para monitorizar a exposição em quatro regiões anatómicas distintitas: olhos, pescoço, tórax e mãos. A aquisição de dados é acompanhada de *feedback* em tempo real com transmissão de informação *wireless* para uma unidade de visualização externa. A deteção da radiação será baseada em fibras óticas cintilantes de plástico (PMMA), cujo coeficiente de atenuação é equivalente aos dos tecidos moles. As fibras serão acopladas a um fotomultiplicador de sílicio, permitindo um *design* compacto e leve. Neste documento é ainda apresentado o plano de trabalho que contém as etapas para o desenvolvimento e avaliação do protótipo durante o ano letivo 2025/2026.

Keywords

Active personal dosimeters, Interventional radiology, Occupational exposure, Real-time monitoring, Scattered radiation.

Abstract

Interventional radiology (IR) is the medical specialty with the highest occupational exposure to ionising radiation, due to its reliance on fluoroscopy. In this modality, scattered radiation is the main contributor to operator dose. Current monitoring systems rely on passive dosimeters that only provide *a posteriori* dose information. This limits the ability to prevent unnecessary exposure.

Active personal dosimeters (APDs) are identified as a complementary solution. These systems provide real-time feedback that can prompt preventive behavioural adjustments. This work reviews the current state of the art of APDs in IR. Commercial solutions and their evolution are presented, along with their operating principles. Reported benefits include general dose reduction, dependent on baseline exposure conditions, and improved team awareness of the radiation received during interventions. Their role as an educational tool for radiation safety practices is also recognised. However, existing devices are costly and limited in anatomical coverage, as they are restricted to chest-level monitoring. In response to these limitations, a new APD prototype is proposed for development during the next academic year. It is designed to monitor exposure at four different anatomical regions: the eyes, neck, thorax and hands. Measurement data will be transmitted wirelessly to an external display, enabling real-time feedback to the users. The sensing unit will be based on plastic (PMMA) scintillation optical fibres, with an attenuation coefficient similar to that of soft tissues. The fibres will be coupled to a silicon photomultiplier for a compact and lightweight design. This document also presents the workplan that will guide the development and evaluation of the prototype during the 2025/2026 academic year.

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List of Abbreviations

ALARA	*As Low As Reasonably Achievable*
APD	Active Personal Dosimeter
FOV	Field of View
FT	Fluoroscopy Time
GUI	Graphical User Interface
ICRP	International Commission on Radiological Protection
IR	Interventional Radiology
KERMA	Kinetic Energy Released in MAtter
LET	Linear Energy Transfer
MCU	Microcontroller
MPPC	Multi-Pixel Photon Counter
OSLD	Optically Stimulated Luminescence Dosimeter
PMMA	Polymethyl Methacrylate
PSOF	Plastic Scintillation Optical Fibre
SiPM	Silicon Photomultiplier
TLD	Thermoluminescent Dosimeter

List of Symbols

D – Absorbed dose

E – Effective dose

E_b – Binding energy of the electron

E_c – Energy of the Compton-scattered photon

E_o – Energy of the incident photon

E_{pe} – Energy of the photoelectron

ε – Deposited energy

ε_{tr} – Transferred energy

H – Equivalent dose

$H_p(d)$ – Dose equivalent in soft tissue at a depth d

K_{col} – Collision kerma

R – Radiant energy

θ – Angle of Compton dispersion

$\bar{W}_{air}$ – Average energy required to produce an ion pair in air

w_R – Radiation weighting factor

w_T – Tissue weighting factor

X – Exposure

Z – Atomic number

Motivation and Contextualization

Interventional radiology refers to image-guided therapeutic procedures performed under real-time X-ray visualization [1]. The number of these procedures has increased over the past two decades worldwide. Interventional radiologists and their teams are the most exposed group of medical radiation workers due to the prolonged use of fluoroscopy and the proximity required during interventions [2,3]. Currently, protection strategies rely on lead garments and structural shielding within the room [4]. Compliance with established occupational dose limits is monitored using passive dosimeters, which are worn either above or below the lead apron. These devices are read *a posteriori*, in the best-case scenario, once a month. As a result, there is no immediate feedback during procedures and exposure incidents cannot be prevented [5,6].

Active personal dosimeters offer a complementary solution by providing real-time auditory and/or visual feedback. The main benefit of these devices in IR is not to prevent exceedance of annual dose limits, which have been extensively studied and are rarely surpassed under standard conditions. Instead, ADPs support the principle of keeping exposure *as low as reasonably achievable* (ALARA) [7–9]. These systems alert personnel to high-dose situations, promoting behavioural adjustments to reduce avoidable exposure. Adjustments may include reducing fluoroscopy time and frame rate, optimizing patient and operator positioning, and improve radiation shielding placement [10].

The proposed project explores the development of an APD system for IR, capable of monitoring radiation exposure at four anatomical regions: eyes, neck, thorax (chest) and hands. The system will transmit dose data wirelessly to an external dashboard for real-time visualization.

Chapter 1. Introduction

1.1 Interventional Radiology

Interventional radiology (IR) is a subspecialty of radiology that utilizes fluoroscopy to perform diagnostic and therapeutic procedures through minimally invasive techniques [11]. Fluoroscopy involves the use of X-rays and an image detection system to visualize internal structures and interventional tools in real-time [12]. The conventional IR equipment configuration is illustrated in Fig. 1. The X-rays are produced in an X-ray tube by converting the kinetic energy of electrons into electromagnetic radiation. It consists of two electrodes enclosed in a vacuum-sealed glass envelope. The cathode serves as the electron source, while the anode acts as the positive target. When heated by an electric current, the cathode emits electrons via thermionic emission. The number of electrons produced depends on the current passing through the filament. A high potential difference applied between the cathode and the anode accelerates the electrons toward the anode. The total incident energy delivered to the anode is proportional to both the voltage and the current [1, 13].

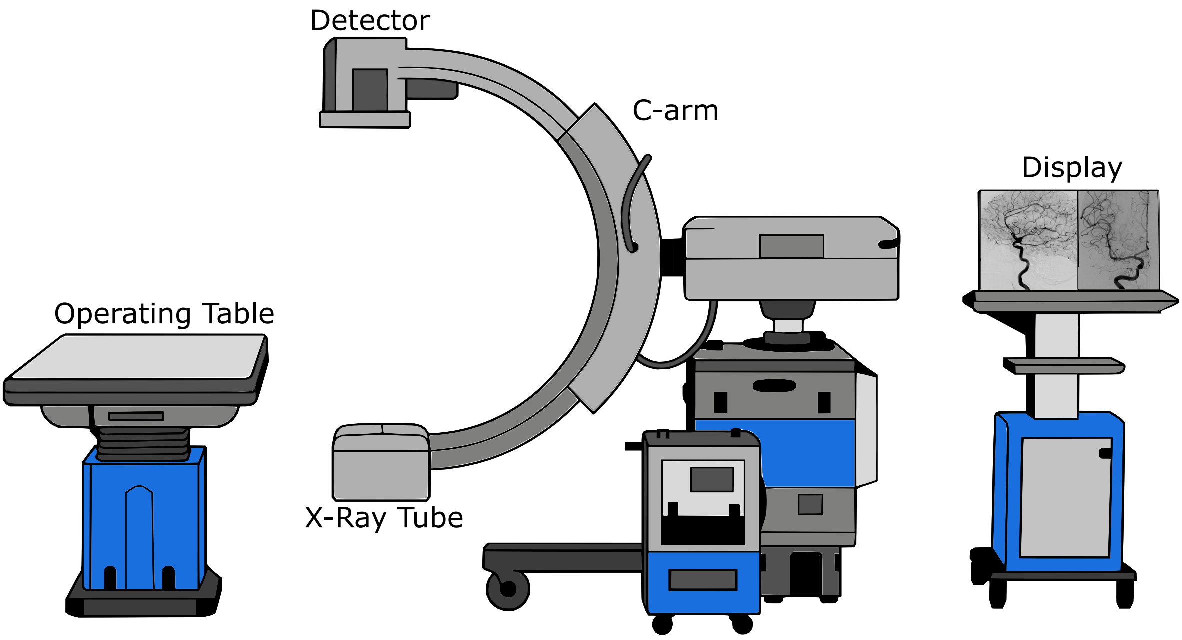


Figure 1: Interventional Radiology Setup. The C-arm is positioned in a posteroanterior projection (standard clinical practice) directing the most intense backscatter towards the floor, protecting above-knee radiosensitive organs [14]. The images shown on the display monitor were reproduced from Refs. [15, 16].

The X-rays transmitted through the patient strike the image detection system, mounted opposing the X-ray tube on a C-arm gantry. In the standard configuration, the X-ray tube is positioned beneath the patient table and the detector above, capitalising on the anisotropic distribution of scattered radiation to minimise occupational exposure [17, 18]. The detector converts incident X-rays into visible light, which is optically transmitted to a camera system and displayed on a monitor to guide interventions in real-time. For continuous fluoroscopy, a steady tube current is supplied and images are acquired at 30 frames per second to maintain the perception of smooth motion [19]. Table 1 outlines common technical parameters used in IR procedures [20].

Table 1. Representative ranges of technical parameters used in IR procedures. Adapted from Ref. [20].

Parameter	Peak high voltage	Tube current	Inherent Al filtration	Cu filtration	Pulse duration	Pulse frequency	Scatter Energy
Range	60–120 kV	5–1000 mA	4.5 mm	0.2–0.9 mm	1–20 ms	1–30 Hz	20–100 keV

1.2 Photon Interaction with Matter

X-rays interact with matter through four physical mechanisms: the photoelectric effect, Compton scattering, Rayleigh dispersion and pair production. The latter two are not relevant in IR [21]. Therefore, only the photoelectric effect and Compton scattering are considered in detail. Both processes are shown schematically in Fig. 2.

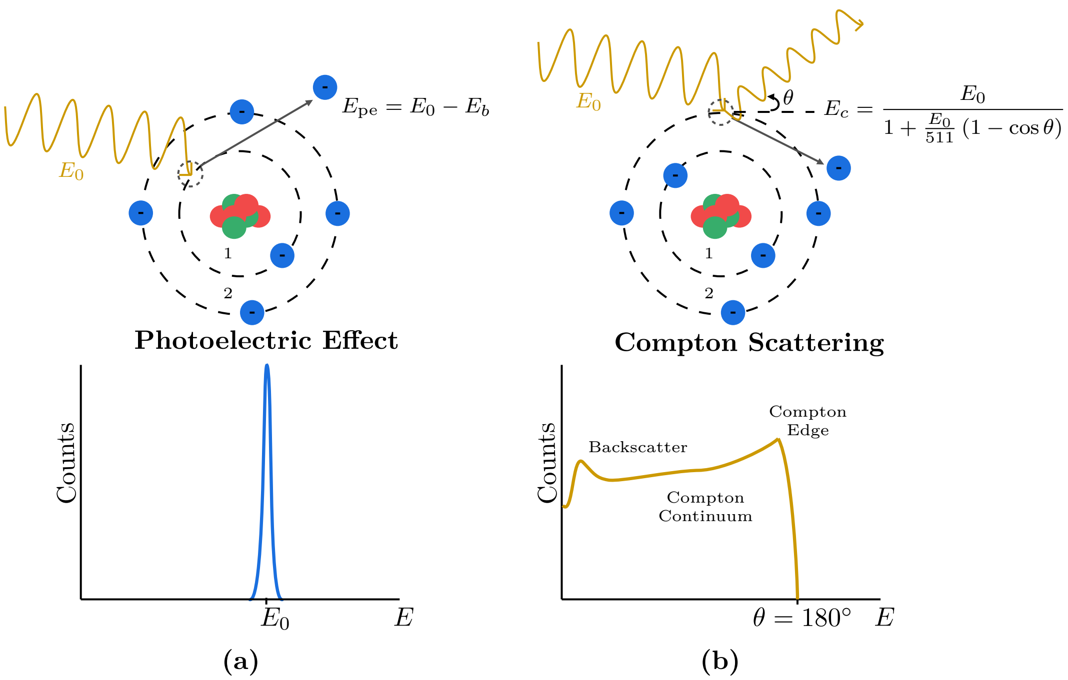


Figure 2: Schematic representation of the photoelectric effect **(a)** and Compton scattering **(b)**, and the resulting energy spectrum.

1.2.1 Photoelectric Effect

The photoelectric effect occurs when an incident photon transfers all of its energy to a bound electron within an atom, resulting in the ejection of the electron from its orbital. This process preferentially involves inner-shell electrons due to their greater electron density. The incident photon energy (E_0) is expended to overcome the binding energy of the electron (E_b), and any remaining energy is converted into the kinetic energy of the ejected photoelectron (E_{pe}), expressed as [1, 13]:

$$E_{\text{pe}} = E_0 - E_b, \text{ unit: [keV]} \tag{1.1}$$

The vacancy is filled by an outer electron, with the release of a characteristic X-ray or an Auger electron. When these emissions are reabsorbed within the detector, the total deposited energy equals E_0, recorded as the photopeak. As the emissions are monoenergetic, the photopeak would be a Dirac delta. However, and as shown in Fig. 2 (a), the peak has a finite width, set by the energy resolution of the detector and is a property of the instrument [21].

The probability of photoelectric interaction is predominant at lower energies (<50 keV). It is also dependent on atomic number, increasing with higher-Z materials, which provides contrast in X-ray imaging. As a result, lower-energy X-ray beams produce higher contrast but also increase photon absorption in the patient, leading to a higher dose. In interventional planning, beam quality is therefore optimised to balance image quality and patient dose, as higher photon energies reduce patient absorption but increase the contribution of Compton scattering, leading to greater noise and poorer contrast [1].

1.2.2 Compton Scattering

Compton scattering is an inelastic interaction in which an incident photon interacts with a free outer-shell electron, transferring part of its energy. As a result, the electron is ejected as a recoil (Compton) electron. In accordance with the conservation of energy and momentum, the photon is deflected at an angle θ, dependent on the energy transferred. The energy of the scattered photon E_c is described as [1]:

$$E_c = \frac{E_0}{1 + \frac{E_0}{511}(1 - \cos\theta)}, \text{ unit: [keV]} \tag{1.2}$$

Within the detector, all scattering angles are permitted, from 0° to 180°. This creates a Compton continuum, bounded by the two extremes. The maximum energy transfer to the electron occurs at $\theta = 180°$, the Compton Edge (CE). At the same angle, a photon scattered in the surrounding material may enter the detector and be absorbed, forming the backscatter peak shown in Fig. 2 (b) [21].

At photon energies typical of diagnostic radiology, the scattered photon retains most of the initial energy and remains sufficiently energetic to penetrate tissue. Within this energy range, Compton scattering predominates over the photoelectric effect in materials such as soft tissue, water and air. The interaction probability is proportional to the physical density of the material, but it is independent of the atomic number due to the uniform electron density across soft tissues [1, 13].

1.3 Dosimetry Fundamentals

1.3.1 Physical Quantities

KERMA

When a beam of indirectly ionizing radiation, such as X-rays, interacts with matter, it transfers energy to charged particles within the medium. The total energy transferred (ε_{tr}) within a volume, V, is defined as [22]:

$$\varepsilon_{tr} = (R_{in})_u - (R_{out})_u^{\text{nonr}} + \sum Q, \text{ unit: [J]} \tag{1.3}$$

where $(R_{\text{in}})_u$ and $(R_{\text{out}})_u$ are the radiant energies of uncharged particles entering and leaving the volume, respectively, excluding secondary radiative emissions. $\sum Q$ represents changes in rest mass energy within V.

The dosimetric quantity *KERMA* (Kinetic Energy Released in MAtter) quantifies the initial kinetic energy transferred to charged particles, per unit mass, regardless of where and how that energy is spent [1, 22]:

$$KERMA = \frac{d\varepsilon_{tr}}{dm}, \text{ unit: } \left[\frac{\text{J}}{\text{kg}}\right] \text{ or [Gy]} \tag{1.4}$$

Absorbed Dose

Absorbed dose (D) incorporates all subsequent energy deposition mechanisms such as radiative losses, e.g. *Bremsstrahlung.* The total deposited energy (ε) includes the energy balance defined in Eq. 1.3, along with all contributions from charged particles, per unit mass at a point of interest. D is used to quantify the effects of radiation and is expressed as [12, 22]:

$$D = \frac{d\varepsilon}{dm}, \text{ unit: [Gy]} \tag{1.5}$$

Exposure

Exposure (X) is the earliest defined dosimetric quantity and applies to photon interactions in air. It measures the amount of electric charge, dQ, produced by ionization in a given mass of dry air [22]:

$$X = \frac{dQ}{dm}, \text{ unit: } \left[\frac{\text{C}}{\text{kg}}\right] \tag{1.6}$$

Exposure is directly related to the collision component of *KERMA*, (K_{col}). K_{col} represents the fraction of energy transferred that is deposited through ionization and excitation, excluding radiative losses. In fact, a connection between X and K_{col} can be established through the quantity $\overline{W}_{\text{air}}$, defined as the mean energy required to produce an ion pair in dry air ($\overline{W}_{\text{air}} = 33.97$ J/C):

$$K_{\text{col, air}} = \overline{W}_{\text{air}} \cdot X = 33.97 \cdot X, \text{ unit: [Gy]} \tag{1.7}$$

In practice, $K_{\text{col, air}}$ is commonly referred to as *air KERMA* and is used to describe the output of an X-ray tube, as it can be measured directly in free air [12].

1.3.2 Protection Quantities

Radiation effects on humans vary due to differences in biological response and are classified as either stochastic or deterministic. Deterministic effects (Fig. 3 (a)), including development of cataracts or skin erythema, have a threshold dose and increase in severity with higher exposure. Stochastic outcomes (Fig. 3 (b)), such as cancer, occur statistically, with the probability increasing with dose but without a threshold. To reflect these probabilistic biological responses, additional weighting factors for radiation type and tissue radiosensitivity must be considered. Thus, two protection-related quantities, equivalent dose (H) and effective dose (E), are derived from absorbed dose [21, 23].

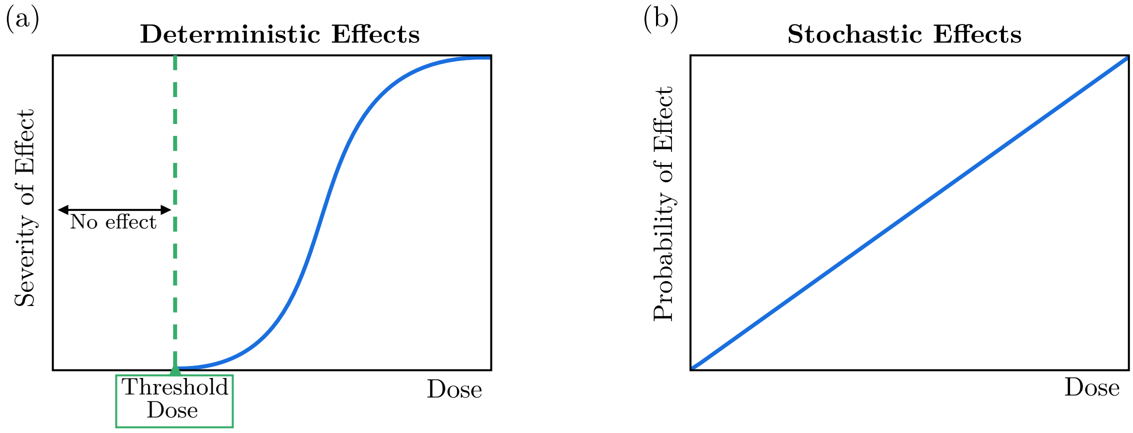


Figure 3: (a) Deterministic effects exhibit a threshold dose below which no effect occurs. Above the threshold, severity increases with dose. Thresholds are expressed in Gy (absorbed dose) as deterministic effects depend only on how much energy is deposited. **(b)** Stochastic effects have no threshold; the probability of occurrence increases with dose, but severity is independent of exposure. The individual either develops the condition or not [23].

Equivalent Dose

The stochastic effects of radiation depend on the type of particles involved, as they interact with matter in distinct ways, exhibiting different patterns of ionization along their paths [22]. This variation is described by the linear energy transfer (LET) function, which represents the rate at which energy is deposited along a particle's trajectory through a medium. Low-LET radiation produces ionization over longer paths, whereas high-LET radiation transfers energy more densely over shorter distances, enhancing its potential for biological damage per unit of absorbed dose [1].

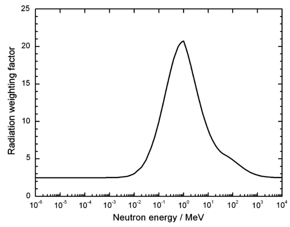
To account for these differences in LET, the International Commission on Radiological Protection (ICRP) introduced the concept of equivalent dose. This measure applies an empirical and dimensionless radiation weighting factor, w_R, to the absorbed dose, to reflect the relative biological effectiveness of different types of radiation [1, 21]:

$$H_T = \sum_R D_{T,R} \cdot w_R, \text{ unit: [Sv]} \tag{1.8}$$

The sievert (Sv) is dimensionally equivalent to joules per kilogram (J/kg) and gray (Gy), but specifically used for radiological protection to distinguish it from the physical quantity D [1].

As shown in Table 2, fast electrons, X-rays and γ-rays have a weighting factor w_R of one, meaning that absorbed dose and equivalent dose are numerically equal [21]. In contrast, α particles have a w_R of 20 due to their higher ionization density. Thus, α particles can produce the same biological outcome as X-rays at only 5% of the absorbed dose [21, 23].

Table 2. Radiation weighting factors. Adapted from Ref. [24].

Radiation type	Radiation weighting factor, w_R
X-rays, γ-rays and Electrons	1
Protons	2
α and Heavy Particles	20
Neutrons	 Continuous function of neutron energy

Effective Dose

Effective dose extends the concept of equivalent dose by incorporating tissue weighting factors, w_T. These factors quantify the relative contribution of specific organs and tissues to the overall radiation detriment, and are represented in Table 3. The values are derived from epidemiological and radiobiological data and reflect averaged population risk, making E a comparative metric rather than an indicator of individual risk. Effective dose is defined as the sum of the equivalent doses received by individual organs and tissues, each weighted by its respective factor [1, 21, 23]:

$$E = \sum_T H_T \cdot w_T, \text{ unit: [Sv]} \tag{1.9}$$

Table 3. Tissue weighting factors for each organ/tissue. Adapted from Ref. [24].

Tissues and Organs	Weighting Factor, w_T	$\sum w_T$
Red bone marrow, Colon, Lung, Stomach, Breast, Remaining tissues*	0.12	0.72
Gonads	0.08	0.08
Bladder, Esophagus, Liver, Thyroid	0.04	0.16
Bone surface, Brain, Salivary glands, Skin	0.01	0.04
	Total	**1.00**

*Adrenals, Extrathoracic region, Gall bladder, Heart, Kidneys, Lymphatic nodes, Muscle, Oral mucosa, Pancreas, Prostate, Small intestine, Spleen, Thymus, Uterus/cervix

1.3.3 Operational Quantity

In operational dosimetry, equivalent and effective dose cannot be measured directly [23]. Personal dosimetry is the applied science of using individual monitoring devices to estimate dose from occupational exposure. These devices are calibrated in terms of the personal dose equivalent, $H_p(d)$, defined as the dose equivalent in soft tissue at a depth d mm beneath a specified point on the body and is expressed in Sv. At 10 mm, $H_p(10)$ is used to estimate effective dose from deeply penetrating radiation and is the principal quantity for assessing whole-body exposure. Dosimeters intended to measure Hp(10) are calibrated using reference radiation qualities defined in ISO 4037-1 [25]. During calibration, the dosimeter is mounted on a polymethyl methacrylate (PMMA) phantom, and the field is characterized in terms of *air KERMA*. For each radiation energy E and angle of incidence α, the International Commission on Radiation Units and Measurements (ICRU) provides conversion coefficients $h_{pK}(10; E, \alpha)$, which relate *air KERMA* to $H_p(10)$ [26]:

$$H_p(10) = h_{pK}(10; E, \alpha) \cdot K_{\text{air}}, \quad [\text{Sv}] \tag{1.10}$$

The same calibration approach applies to the assessment of equivalent dose, for which shallower depths are used. At 0.07 mm, $H_p(0.07)$ is used to assess dose to the skin from weakly penetrating radiation, and at 3 mm, $H_p(3)$ has been proposed specifically for the lens of the eye [23].

1.4 Occupational Exposure

As previously mentioned, at the photon energies typical of fluoroscopy (recall Table 1), the Compton effect predominates, degrading image quality and generating scattered radiation throughout the room [1, 27]. This stray radiation is the main source of occupational exposure for staff [28]. From a dosimetric perspective, characterising the IR settings is inherently challenging due to the variability of parameters that influence radiation output and scatter distribution [29]. In general, exposure to staff is proportional to the dose received by the patient [17]. Tube voltage and current are adjusted based on patient size and anatomy during procedure planning, which in turn influence the amount of scatter produced [29, 30]. In addition, FOV and beam filtration also play a role, though their effects are not linear and highly dependent on scatter geometry [31]. Beyond equipment settings, procedural complexity and operator technique contribute to variations in staff exposure. Advanced or prolonged interventions typically require longer fluoroscopy times, more image acquisitions, and non-standard projections [29]. Operator experience has been shown to significantly reduce exposure through more efficient techniques, reduced reliance on fluoroscopy, and better positioning practices [32].

To illustrate scatter behaviour in a real clinical context, Fig. 4 presents an isodose map, at chest height, generated during a fluoroscopically guided hip replacement procedure, and highlights the locations typically occupied by staff [33]. The image reveals an asymmetric scatter pattern, with notably higher intensities recorded opposite to the position of the C-arm. The referenced study also reported lower dose levels on the side opposite the patient table, where equipment provides partial shielding. Although the inverse square law does not perfectly apply in the clinical environment [34] due to scattering within the patient and surrounding structures, a general decrease in dose rate with increasing distance was still evident [33]. This reaffirms the protective effect of physical separation from the source [35].

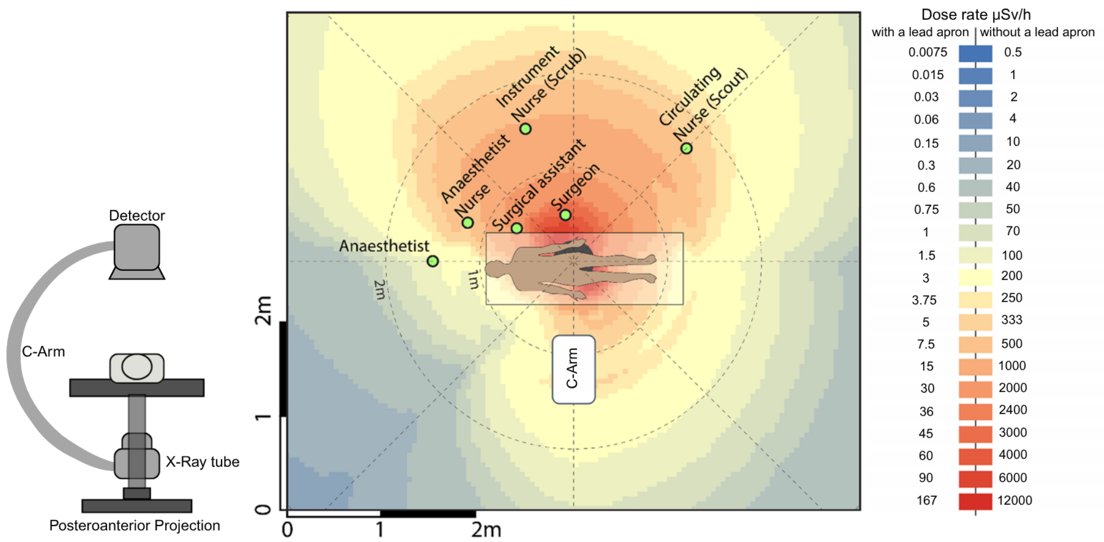


Figure 4: Isodose map of scatter radiation during a fluoroscopically guided hip replacement procedure. Acquisition parameters included 77 kVp, 20.6 mA, 32.1 s of fluoroscopy time, and a pulse rate of 15 fps. Dose rates were measured using an active dosimeter at multiple positions around the phantom and linearly interpolated to generate the map. Values incorporating shielding (right scale) were estimated using theoretical models. Measurements are documented at chest level. The image was adapted from Ref. [33]. Although the C-arm configuration was not explicitly reported, a posteroanterior projection (left icon) was inferred based on findings from Ref. [34].

Protection against cumulative exposure requires consistent application of shielding practices [4]. Shielding is implemented at multiple levels, including structural room design, fixed barriers, and mobile shields, which contribute to protecting helping staff [36] and sensitive areas such as the head and eyes, respectively [37]. However, personal protective equipment remains the most reliable and controllable form of protection. According to ICRP recommendations, all staff should wear lead aprons, along with thyroid shields, during fluoroscopic procedures [4, 37]. Most aprons have a lead thickness range of 0.35 to 0.5 mm, with 0.5 mm reducing scattered radiation by over 90% [4]. To be effective, aprons must fit properly, and facilities should be encouraged to invest in a variety of sizes to ensure proper coverage and comfort [36,38]. Thyroid shields are manufactured with a lead equivalence of 0.5 mm and have been reported to reduce exposure to the gland by approximately 50% [39]. Lead glasses are also available but remain underutilised. As the second most effective protective measure after ceiling-suspended shields, these glasses are available in various frame designs and thicknesses ranging from 0.35 to 0.75 mm.

The effectiveness of these protective measures is ultimately verified against the annual occupational dose limits recommended by the ICRP, summarised in Table 4. These limits are not an absolute safety threshold but rather boundaries beyond which cumulative exposures are no longer considered justified [23]. For instance, the E limit for radiation workers is set at 20 mSv per year, averaged over five years, with an upper limit of 50 mSv in any single year [24].

Table 4. ICRP occupational dose limits. Adapted from Refs. [24, 40].

Protection Quantity		Annual Permissible Dose
Effective Dose		20 mSv per year averaged over 5 years (no single year exceeding 50 mSv)
Equivalent Dose	Lens of the Eyes	20 mSv
	Thyroid Gland	300 mSv
	Individual Organs (excluding the eye), Skin and Extremity	500 mSv

As previously introduced, currently, compliance with occupational dose limits in IR is verified through routine monitoring using passive dosimeters worn by staff over predefined time intervals. The most common devices are thermoluminescent dosimeters and optically stimulated luminescence dosimeters. Both operate by trapping energy when exposed to ionising radiation, which is later released as light when stimulated under controlled conditions. The intensity of the emitted light is proportional to the amount of absorbed radiation [12]. These devices measure the previously discussed quantity $H_p(10)$. While $H_p(10)$ can serve as an approximation of effective dose under conditions of uniform whole-body exposure, this does not reflect the typical environment in IR, as described [41]. When a single dosimeter is used, placement beneath the lead apron tends to underestimate E due to shielding, while positioning it above may overestimate it [29]. To improve accuracy, the ICRP recommends a dual dosimetry approach. In this method, one dosimeter is worn underneath the protective apron between the waist and chest to record the attenuated exposure to shielded tissues (H^u). The second device is placed at collar level outside the apron to monitor exposure to unshielded regions (H^o). An effective dose estimate is then calculated by applying a linear combination of these two measurements [42]:

$$E = \alpha H^u + \beta H^o, \text{ unit: [Sv]} \tag{1.11}$$

However, there is currently no global consensus on the value of both weighting factors, α and β [42]. Over 14 different algorithms have been reported in the literature, most of which overestimate E. Despite this variability, double dosimetry remains more accurate than single-dosimeter methods [43].

The major limitation of passive dosimeters lies in their inability to provide dose information in real-time. These devices are typically read at monthly intervals [6], which prevents identification of excessive exposures or deviations from standard operating conditions in time. As a result, opportunities for real-time corrective action are lost, and staff may remain unaware of elevated exposure levels until the data can be reviewed retrospectively [5, 44]. In this context, active personal dosimeters (APDs) have emerged as a complementary solution. APDs are not intended to replace passive dosimeters, which remain the standard for legal dose documentation. By providing real-time feedback on dose rate, APDs support adherence to the ALARA principle by enabling immediate adjustments to operator behaviour or shielding practices [8, 9].

1.5 Real-Time Dosimetry

The concept of real-time radiation monitoring in occupational settings was first established through the use of APDs, which have been widely adopted in the nuclear industry as a standard safety practice [45]. However, it was only within the last 15 years that their use in IR began to be critically evaluated [8,36,45,46]. As previously discussed, the IR environment is characterised by pulsed radiation, lower photon energies, and predominantly scattered fields, which differ significantly from the conditions under which the APDs available at the time were originally developed [46]. Although several studies [20, 45, 46] explored the possibility of adapting existing devices for IR, it was Sánchez *et al.* [8], working as part of an early development initiative at Philips, who introduced the first APD specifically engineered for interventional use. Unlike existing APDs that provided exclusively auditory feedback, this prototype included a visual interface displaying real-time dose data to staff in the procedure room. It became the first generation of the DoseAware system, which started being distributed commercially in 2010 [47]. Later, Philips and Unfors RaySafe signed a joint development agreement in 2012 and, in 2017, the third-generation system, RaySafe i3, was introduced at the European Congress of Radiology [48].

The device features a silicon solid-state diode [49], integrated into an invididual wearable bagde with wireless capabilities, transmitting scattered dose information to a data collection unit at a one second rate. Detected dose values are displayed on an in-room monitor as both numerical data and a bar graph. Each badge is represented individually, with exposure levels indicated using a colour-coded scheme (green, yellow, and red) corresponding to increasing dose rates [8]. The operational photon energy range of the system is N40–N150 for X- and γ-radiation. The relative measurement uncertainty is approximately 10% for dose rates in the range of 40–150 μSv/h and 20% for dose rates between 150–300 μSv/h. The system has been shown to correlate well with legal-for-record passive dosimeters, typically overestimating $H_\mathrm{p}(10)$ by 5–15%.

The collaboration between Philips and RaySafe enabled the integration of badge-derived data into dedicated software platforms, such as Dose Viewer and Dose Manager [50]. These platforms support retrospective analysis by allowing users to access, export, and manage individual dose histories, as well as aggregate exposure data across teams or departments. RaySafe i3 is currently priced at around 27,000€ [51], which raises concerns about cost. Individual APD units are expensive, making it difficult for most imaging departments to provide them for all staff. In addition, current systems do not support extremity monitoring, such as for the eyes or hands [52].

1.5.1 Reported Benefits

The development and clinical integration of the APDs described above enabled structured evaluation of their practical impact in IR. Since their introduction, a number of studies have compared conditions with and without real-time feedback [10, 44, 49, 53–61]. A common finding is a general decrease in staff dose when feedback is available. This effect appears across various professional roles, though the degree of reduction can vary. In a study led by Baumann *et al* [56] dose reduction was observed in almost every staff group [56]. Further analysis reveals a pattern seen across multiple studies, where the most significant reductions are recorded in the main radiologist, who is positioned closest to the patient. In some studies, this group was the only one to show a statistically significant change in radiation exposure [49, 56–58]. This may be explained by findings from Baumgartner *et al* [49], which showed that procedural contexts associated with higher radiation exposure tend to demonstrate the most pronounced reductions following the introduction of APDs [49]. This is consistent with reports that the effectiveness of APDs depends on baseline conditions. Operators who initially demonstrated poor radiation safety habits were able to significantly reduce their dose after the introduction of APDs through simple corrective actions. In contrast, those already practising good protection strategies had less room to improve and showed minimal change [53, 57, 60]. These baseline variations are also associated with the operator's level of experience, as discussed in section 1.4, and were recently studied by Koch *et al* [54]. A pronounced decrease in dose was registered among the least experiences group, while operators with over ten years of experience show, in this case, even a slight increase in dose. Experienced operators may rely on ingrained habits and confidence in their technique, whereas less experienced users appear to adjust more readily when confronted with live feedback [54]. It has been reported that younger operators tend to show less concern for radiation safety [62]. These results suggest that APDs may be especially valuable as educational tools for junior staff, helping them build good radiation safety practices [54].

Consistent with these findings, James *et al* [60], reported a reduction in red events over time. They argued that, while these events serve as prompts for immediate dose adjustment, they also function as a reflective tool for operators. This enables professionals to identify specific maneuvers, projection angles, and procedural choices associated with increased radiation exposure. Consequently, such practices can be replaced by lower-exposure alternatives and operators can adopt safer procedural habits more permanently. Ultimately, it also contributes to a reduction in radiation dose to the patient [60, 61].

Other staff members, including nurses and technicians, also benefited from dose reductions [53–56, 58, 60]. In these cases, it was primarily attributed to individual awareness and simple adjustments, such as changes in positioning. Some studies further emphasised the collaborative value of these systems, noting that when all staff members wear APDs with visible displays, teams can collectively monitor exposure. This practice creates an environment of mutual accountability, where individuals not only adjust their own behaviour but also remain aware of their colleagues' exposure, promoting safer team dynamics [56, 57, 60].

Chapter 2. DORI - Hardware Development

This chapter describes the development of the DORI (*DOsímetro para Radiologia de Intervenção*) prototype, designed in response to the previously identified limitations. The conceptual design of the system is shown in Fig. 5.

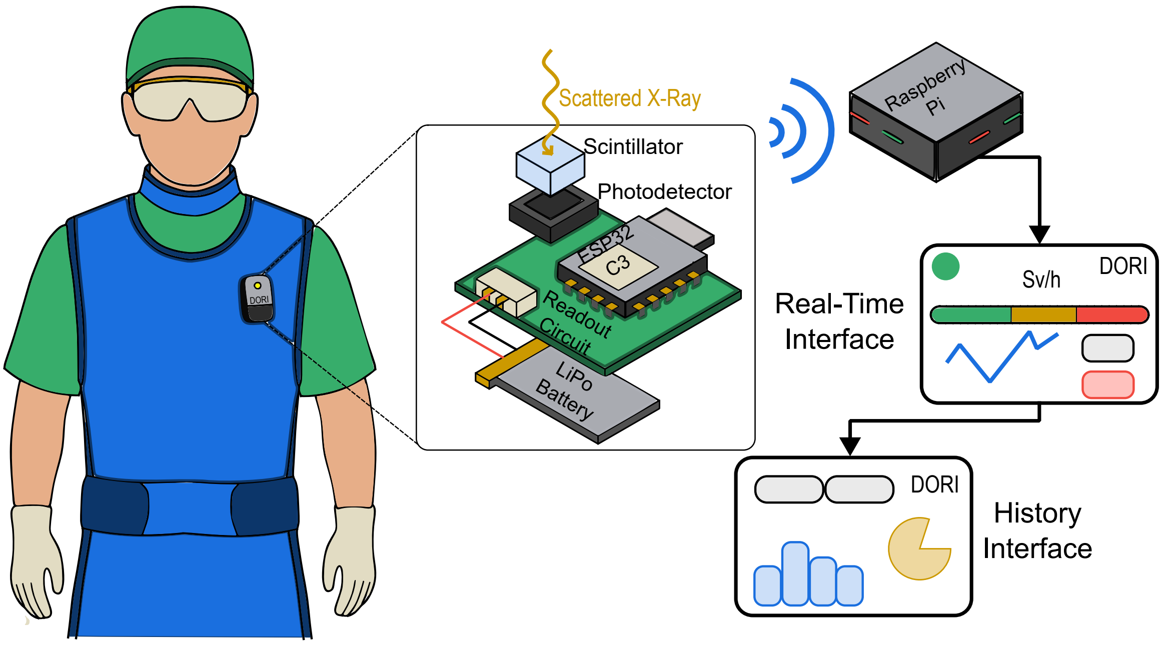


Figure 5: Conceptual design of the DORI system. The wearable unit (exploded view) is worn over the protective apron and comprises the scintillator, photodetector, readout circuit, MCU and LiPo battery. Data are sent to a Raspberry Pi, which displays the real-time and history interfaces.

DORI consists of a compact wearable unit, worn over the protective apron, which integrates the full detection, pulse processing and transmission chain. Each stage is described in detail in the following sections. In general terms, scattered radiation interacts with a scintillator, and the resulting light is converted into an electrical signal by a photodetector. DORI operates in pulse mode, where each interaction is registered as an individual event. The corresponding pulse is conditioned by a readout board, which preserves the amplitude and thereby the energy deposited in each event. By accounting for the energy of individual events, rather than integrating total charge, the accuracy of the dose calculation can be improved. The target energy range is 20 to 100 keV, corresponding to the energies of the scattered field in IR (recall Table 1).

The chain is controlled by a microcontroller unit (MCU) with wireless communication capabilities. The ESP32-C3 was selected for being compact, lightweight and low-cost. It supports 2.4 GHz Wi-Fi and Bluetooth® Low Energy (BLE) 5 [63], the latter being the adopted communication protocol. Unlike Wi-Fi, BLE establishes a direct link with the receiver and requires no external router, reducing system complexity. In addition, BLE 5 offers extended range and increased data throughput relative to previous versions, meeting the requirements of continuous dose transmission. The unit is powered by a rechargeable lithium polymer (LiPo) battery, the AKYGA LP103040 [64], whose 1200 mAh capacity supports uninterrupted operation over extended periods. The selected Seeed Studio XIAO ESP32-C3 board already incorporates a charging circuit for the battery via its USB-C port [65]. Accordingly, the front-end electronics operate at a 3.3 V logic level.

To preserve battery life, the wearable unit does not compute dose. The MCU accumulates the amplitude information of the detected events and transmits it via BLE at a one-second rate to a Raspberry Pi. Since the Raspberry Pi is permanently powered, it handles all data processing, from pulse amplitude to energy and, ultimately, to dose. A real-time interface then displays the dose rate with a complementary colour-coded scale, updated at the same one-second rate. The scale is replicated on an LED integrated in the wearable unit, with the corresponding colour level sent back to the badge via BLE. All measured data are also stored in an internal database, from which a second interface builds retrospective summaries of the recorded exposure.

2.1 Scintillator

The detection of the scattered radiation is based on a plastic scintillator (PS). PSs are organic materials that emit photons in the near ultraviolet to visible range when excited by ionising radiation. The selected scintillator is a $5 \times 5 \times 5$ mm^3 BC-408 cube. BC-408 consists of a polyvinyltoluene base doped with fluorophore compounds that shift the emission to a peak wavelength of 425 nm [66].

The main advantage of PSs is their tissue equivalence. Their Z_{eff} is also lower than that of the materials used in passive dosimeters or in solid-state diodes. The energy absorbed in the scintillator therefore approximates the dose to soft tissue without requiring further corrections [67]. However, as a drawback, the same tissue equivalence implies that, as in soft tissue, Compton scattering predominates over the photoelectric effect at the photon energies of the scattered field. From an energy calibration perspective, the acquired spectrum reduces to a Compton continuum without photopeaks, and the standard identification of gamma-emitting radionuclides is not possible (recall Section 1.2). Consequently, the energy resolution of these detectors is poor. The only identifiable spectral feature is the Compton edge (CE), corresponding to the maximum energy transfer at a scattering angle of $\theta = 180°$, whose energy is well known for a given source. Nonetheless, owing to the poor energy resolution at the energies of interest, the CE may not coincide with the channel of maximum counts. To resolve this ambiguity, Compton coincidence techniques are employed, in which the backscattered photon from a CE event is detected in coincidence with a reference detector. However, these have primarily been demonstrated using ^{22}Na and ^{137}Cs, whose CEs lie outside the range relevant to this work [68].

Such methods also assume a proportional response, meaning a light output that scales linearly with the deposited energy. However, below approximately 125 keV, the light output of PSs is non-proportional due to ionisation quenching. Slow (lower energy) electrons ionise the medium more densely, and part of the excitation is dissipated without the emission of light [69]. The Compton electrons produced in the PS in this work fall below this threshold. The pulse height of the detected events is therefore not a linear measure of the deposited energy, and the response must instead be characterised locally, with any calibration valid only within the region where it is established.

2.2 Photodetector

To detect the scintillation light and convert it into an electrical signal, a SiPM from Broadcom (AFBR-S4N66P014M) was optically coupled to the scintillator. Its 6×6 mm^2 active area matches the PS cross section, maximizing light collection. Additionally, the photon detection efficiency of the SiPM peaks at 420 nm, close to the peak emission of BC-408 [70].

A SiPM consists of a parallel array of avalanche photodiodes operated in Geiger mode. Each cell is reverse biased above its breakdown voltage, so a single absorbed photon is sufficient to trigger a self-sustaining avalanche. The photons produced by a scintillation event fire many pixels simultaneously, each of which contributes one identical pulse. The summed output current is proportional to the number of fired cells, and therefore to the energy deposited in the PS [21]. The number of available microcells (22 428) fulfills the requirements given the photons produced per scintillation event. The SiPM has a breakdown voltage of 33 V and supports overvoltages of up to 16 V [70].

2.2.1 SiPM Power Supply

The SiPM operating voltage was set to 37 V. Although gain increases with overvoltage, so do the dark count rate and crosstalk probability, which in excess mask the low-energy end of the spectrum. For the 20 to 100 keV range, an overvoltage of 4 V provides enough gain for these events to sit above noise levels. The bias is generated by a MAX1932 and its output is programmed through a Serial Peripheral Interface (SPI) by the MCU, which writes an eight-bit code to the internal DAC of the power supply [71].

A fixed 37 V output does not account for the variation of the breakdown voltage with temperature, which increases at 30 mV/°C. Since gain depends on the overvoltage, a constant bias means the gain decreases as temperature rises. The same deposited energy then produces a different pulse amplitude, affecting the energy calibration and the derived dose. To compensate for this, an NTC thermistor was added to the feedback network, as recommended in the MAX1932 datasheet and shown in Fig. 6 [70].

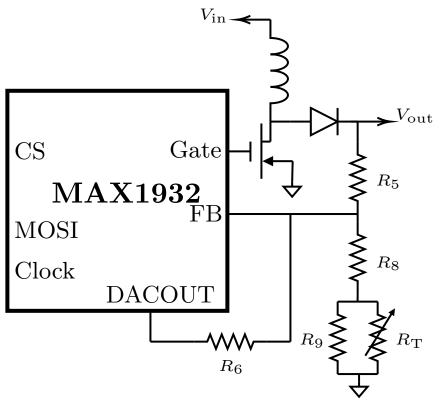


Figure 6: MAX1932 feedback network with the NTC thermistor R_T for temperature compensation of the SiPM bias. Pin labels on the left correspond to SPI connections.

The total resistance between the FB node and ground is:

$$R_{\text{total}}(T) = R_8 + \frac{R_9\, R_T(T)}{R_9 + R_T(T)}, \tag{2.1}$$

The output voltage is therefore:

$$V_{\mathrm{OUT}} = 1.25\left(1 + \frac{R_5}{R_{\mathrm{total}}(T)}\right) + \frac{R_5}{R_6}\left(1.25 - V_{\mathrm{DAC}}\right), \tag{2.2}$$

where V_{DAC} is the DAC output voltage. The compensation condition requires the derivative of Eq. 2.2 to equal the temperature coefficient of the breakdown voltage over the operating range of 18 to 25 °C:

$$\frac{dV_{\mathrm{OUT}}}{dT} = \frac{1.25\, R_5}{R_{\mathrm{total}}^2}\left(\frac{R_9}{R_9 + R_T}\right)^2 \frac{\beta}{T^2} R_T = 30\ \mathrm{mV}/^\circ\mathrm{C}. \tag{2.3}$$

Solving this condition together with Eqs. 2.1 and 2.2 and the component selection equations of the datasheet yields $R_5 = 91\ \mathrm{k\Omega}$, $R_6 = 27\ \mathrm{k\Omega}$, $R_8 = 2.4\ \mathrm{k\Omega}$, and $R_9 = 910\ \Omega$.

2.3 Front-End Electronics

The front-end electronics perform the pulse-mode readout of the SiPM response. The architecture developed for this purpose is represented by the simplified block diagram in Fig. 7, and includes three stages. Pulse shaping integrates and filters the SiPM current into a voltage pulse whose amplitude is proportional to the collected charge. Pulse discrimination then compares this pulse against a reference threshold, rejecting noise, and retains its peak amplitude. Pulse sampling digitises the held value and transmits it to the MCU. Each stage is detailed in the following sections.

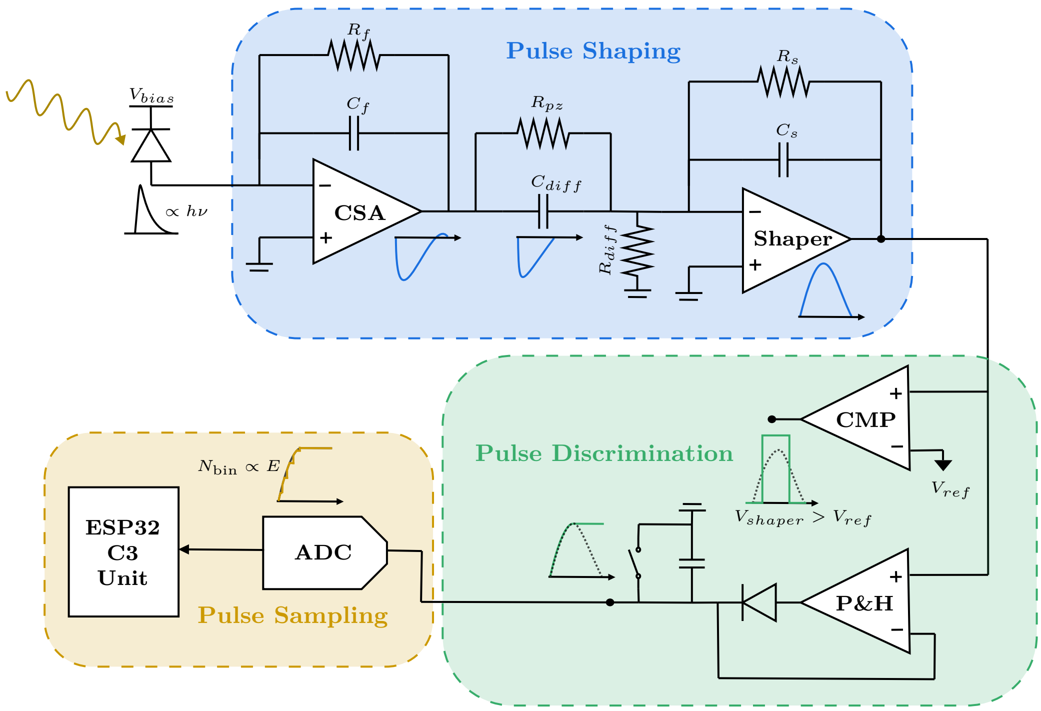


Figure 7: Architecture of the DORI system.

The front-end electronics were assembled on a printed circuit board (PCB), which also integrates the ESP32-C3, the MAX1932 bias supply and the SiPM described previously. The DORI-test board is shown in Fig. **??**.

2.3.1 Pulse Shaping

The current pulse produced by the SiPM delivers a quantity of charge, Q, that must be converted into a measurable voltage signal in the preamplifier [72]:

$$Q = \int I(t)\, dt, \quad \text{unit: [C]} \tag{2.4}$$

To achieve this, a Charge-Sensitive Amplifier (CSA) was implemented. The CSA is widely used as a preamplifier since the rise time of the output pulse depends only on the charge collection time in the detector, independent of both the detector and the preamplifier input capacitances [21]. It is an inverting integrator configuration designed around a low-noise operational amplifier, such as the LTC6228 [73], that was chosen accordingly. The feedback loop is closed by a capacitor C_f in parallel with a resistor R_f [21, 74].

In this circuit, the current flows through C_f, allowing it to store charge and producing an output voltage proportional to Q. R_f acts as a reset mechanism, providing a discharge path for C_f, in order for new pulses to be detected. The resulting output voltage can be written as [72, 75]:

$$V_{\text{out}} = -\frac{Q}{C_f}\, e^{-t/\tau_f}, \quad \text{unit: [V]} \tag{2.5}$$

where $\tau_f = R_f C_f$ is the discharge time constant. For total charge integration, this constant must be longer than the duration of the SiPM current pulse.

The charge gain of the CSA, G, depends exclusively on C_f and is therefore [74]:

$$G = \frac{V_{\text{out}}}{Q} = -\frac{1}{C_f}, \quad \text{unit: [V/C]} \tag{2.6}$$

To preserve the pulse amplitude, which carries the energy information of the detected event, additional signal conditioning stages were implemented. Given the high event count rate expected in IR, a Semi-Gaussian shaper was adopted in order to prevent baseline drift and relax the timing demands of the following stages. It consists of a CR differentiator followed by n RC integrator stages. Increasing n produces a more symmetric output pulse, which returns to baseline faster but introduces a longer peaking time [21]. For this work, a simple CR-RC ($n = 1$) network was implemented, with equal differentiation and integration time constants.

The finite decay time constant of the CSA causes an undershoot at the shaper output [76]. If the signal has not recovered before the next event arrives, the subsequent pulse is superimposed on this undershoot, reducing its measured amplitude. To mitigate this effect, a Pole-Zero Cancellation (PZC) configuration was implemented. C_{diff} and R_{diff} form the CR differentiation network. A resistor R_{pz}, placed in parallel with C_{diff}, introduces a zero that compensates the pole produced by the CSA. Proper cancellation occurs when the following condition is satisfied [21, 77, 78]:

$$R_{\text{pz}}\, C_{\text{diff}} = R_f\, C_f \tag{2.7}$$

The RC stage was implemented as an inverting low-pass filter, similar to the CSA configuration, with a feedback capacitor C_s in parallel with a resistor R_s. As in the CSA, the gain depends on the feedback capacitor, so the value of C_s controls the gain of the stage. The value of R_s is then chosen so that $R_s C_s$ equals the common time constant fixed by the previous stages.

The gain of the front-end electronics was set empirically. As mentioned in Section 2.2.1, the SiPM bias was fixed to keep the dark count rate at a low amplitude without masking the low-energy end of the spectrum. The gain must be high enough that real pulses are resolved above this level. It must also be low enough that the full energy range of interest, up to 100 keV, remains below saturation. The gain therefore places the spectrum in the lower portion of the available 3.3V amplitude range. Within this portion, enough channels must remain available to sample these energies. Following Knoll [21], a spectral feature is adequately sampled when its FWHM spans at least five channels. The reference sources available for this work are limited in the relevant energy range (see Annex **??**). Of these, ^{241}Am is the only source that produces a resolvable photopeak in BC-408, at 59.5 keV. The reported energy resolution of BC-408 at this energy is 30 to 50% [68, 79, 80]. The narrowest case, 30%, corresponds to a FWHM of 17.9 keV and sets the more demanding limit. Five channels across this FWHM give a channel width of 3.6 keV, and therefore a minimum of 28 channels up to 100 keV. This is a lower bound on the channel count, below which the photopeak is undersampled and energy information is lost. However, it is worth noting that such a small number of channels would be impractical for spectral characterisation, and therefore for energy calibration.

The ^{241}Am source was nevertheless used to guide the gain. The pulses generated by the source were observed directly on the oscilloscope, and the gain was adjusted so that they did not saturate the op-amps. It was set to remain within an amplitude range that the ADC samples with a reasonable number of channels, leaving room for the higher energies up to 100 keV. Ultimately, the gain was chosen to give finer binning while still satisfying the condition imposed by the most limiting factor, the resolution of BC-408. The resulting component values of the pulse shaping chain are listed in Table 5.

Table 5. Component values of the pulse shaping chain.

CSA		**PZC**		**Shaper**	
Parameter	**Value**	**Parameter**	**Value**	**Parameter**	**Value**
R_f	4.7 kΩ	R_{diff}	4.7 kΩ	R_s	10 kΩ
		R_{pz}	4.7 kΩ		
C_f	100 pF	C_{pz}	100 pF	C_s	47 pF
τ	470 ns		470 ns		470 ns

The transient response of the shaping chain to a single ^{241}Am event is shown in Fig. 8. The associated time metrics, extracted over the full set of acquired waveforms, are summarised in Table 6.

The integration at the CSA produces a negative pulse, since the charge is collected onto C_f in an inverting configuration, and this polarity is preserved through the differentiating network. The inverting topology adopted for the last integrating stage restores a positive output at the shaper, as required by the single-positive-supply operation of the subsequent stages. A residual undershoot is nevertheless observed at every stage, originating from the CSA. Its relative magnitude is not increased along the chain, which would be the case in the absence of R_{pz}.

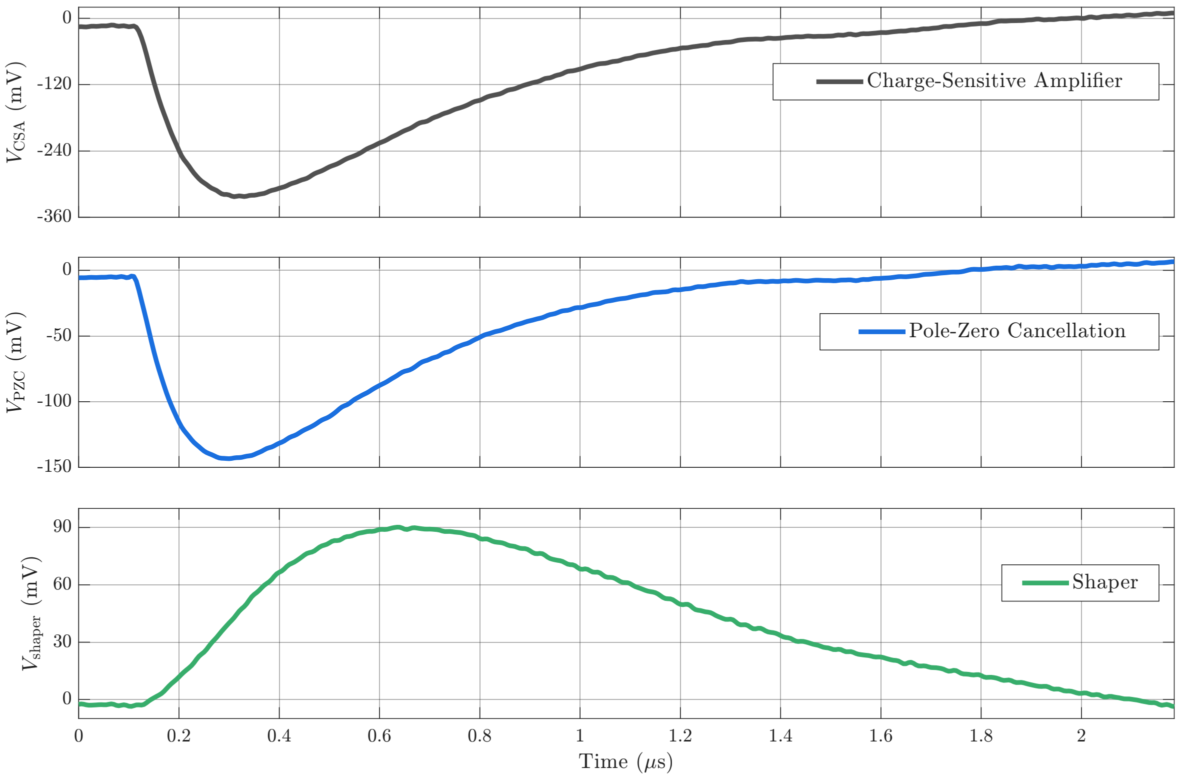


Figure 8: Caption.

Table 6. Timing metrics of the pulse shaping chain at each stage.

	CSA	**PZC**	**Shaper**
Amplitude range (mV)	191–861	87–430	61–407
Peaking time (ns)	147 ± 28	135 ± 25	387 ± 50
Width at 90% of peak (ns)	208 ± 167	192 ± 141	400 ± 154
Total shaping time (μs)	1.42 ± 0.09	1.26 ± 0.07	1.96 ± 0.09

The gain at the final stage was decreased, so that events of this energy are sampled towards the lower end of the available bins, as discussed. The main effect of the shaping is the redistribution of the signal in time. The semi-Gaussian output is broader and the time taken to reach the maximum is increased from 147 ± 28 ns at the CSA output to 387 ± 50 ns at the shaper, even though the latter has a lower amplitude. The interval over which the signal remains within 90% of its maximum is doubled, from 208 ± 167 ns to 400 ± 154 ns. These effects are intended, since the amplitude is retained through comparator logic, and a maximum that is both delayed and sustained reduces the timing demands imposed on both the hardware and firmware. Within a broad range of amplitudes, the total shaping time is consistent across events, as it is set by the time constants selected for this network. At the shaper output it is 1.96 ± 0.09 μs, and this is the pulse the subsequent stages discriminate and sample.

2.3.2 Pulse Discrimination

Pulse discrimination distinguishes real X-ray events from electronic noise and retains the amplitude of those accepted for digitisation. As mentioned above, it operates on a comparator logic. The LT1711 [**?**] was selected for this application, a fast comparator operated in a non-inverting configuration. As shown in Fig. 7, the signal from the shaper is applied to the non-inverting input, while a fixed reference voltage, V_{ref}, is applied to the inverting input. The reference is derived from the 3.3 V supply rail through a resistive divider:

$$V_{ref} = \frac{R_2}{R_1 + R_2} \cdot 3.3, \text{ unit: [V]} \tag{2.8}$$

where R_1 is the resistor connecting the node to the supply, and R_2 the resistor between the divider node and ground.

The threshold was set to 10 mV. This value was established from the comparator input acquired under dark count conditions of the SiPM. The amplitudes of these events lie predominantly below 10 mV, so a threshold placed at this level rejects most spurious triggers without suppressing the low-energy end of the spectrum. The condition is satisfied by $R_1 = 33$ kΩ and $R_2 = 100$ Ω.

A dedicated pin of the ESP32-C3 is connected to the logic output of the comparator and configured as an interrupt. Whenever the input exceeds V_{ref}, the comparator output switches to the logic high level. The pin is monitored continuously by dedicated hardware within the MCU, so the transition is registered as soon as it occurs. Detection initiates the readout loop, and the interrupt is detached for its duration, so that no further triggers are accepted while the event is being processed. It is reattached once the loop has terminated.

2.3.3 Pulse Sampling

2.4 Validation of the Front-End Electronics

Prior to energy calibration of the plastic scintillator detector, the readout electronics were validated using a $5 \times 5 \times 10$ mm CsI(Tl) crystal. The mentioned amplitude sinking in the P&H circuit is inconsequential if constant, but an amplitude-dependent drop would introduce nonlinearities into the sampling network and would require characterisation or correction. An inorganic scintillator is appropriate for this validation due to its higher density and effective atomic number, which favour photoelectric absorption in the energy range of interest. This produces well-defined photopeaks at energies characteristic of each source. Since scintillation light yield in CsI(Tl) is linear with deposited energy, a linear channel-to-energy mapping is expected, and any significant deviation would be attributable to the front-end electronics [21].

The hardware gain was adjusted to accommodate the higher light output of CsI(Tl), and spectra were acquired using the three radioactive sources described previously. The resulting pulse height spectra are shown in (Fig. 9 (a)). All expected spectral features are resolved for each source.

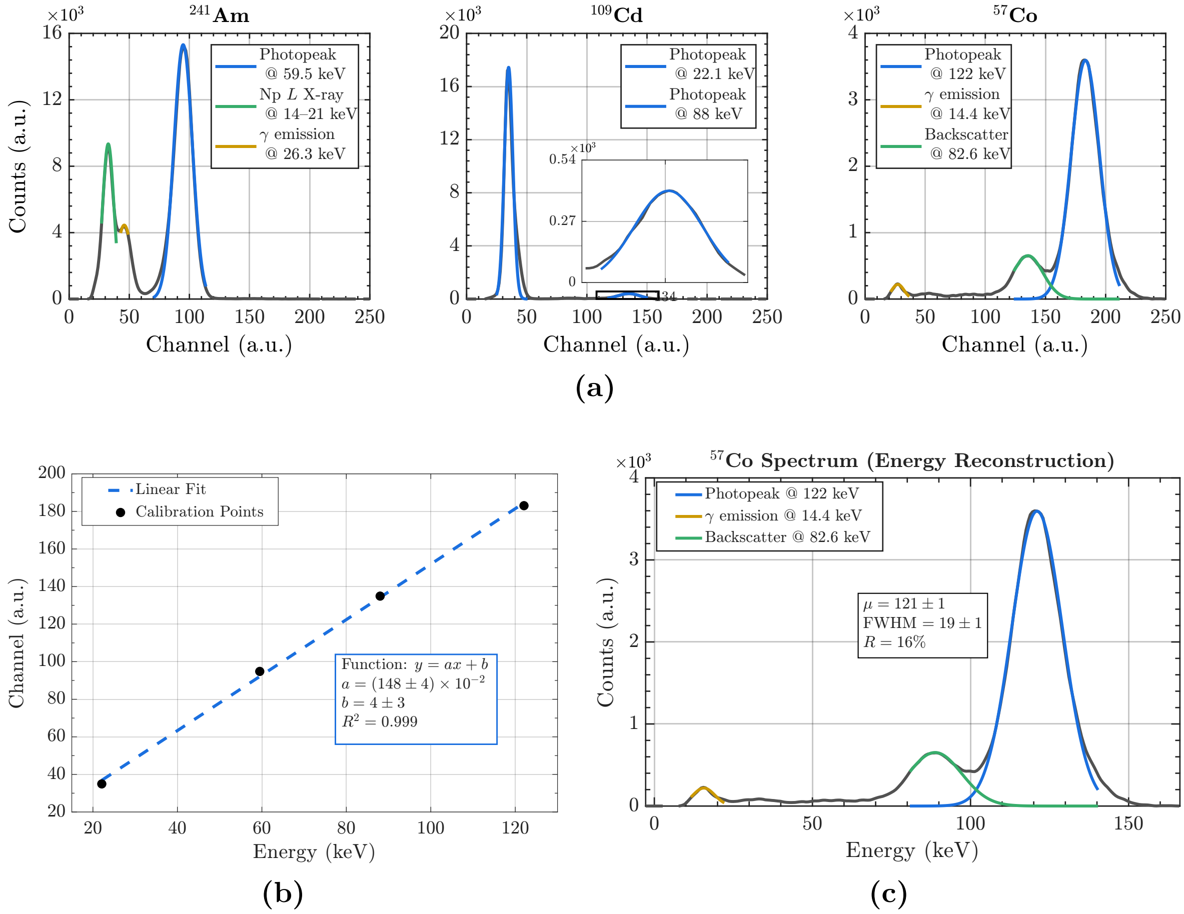


Figure 9: **(a)** Pulse height spectra acquired with the CsI(Tl) crystal for the three calibration sources. Identified spectral features are labelled. **(b)** Channel-to-energy calibration curve with linear fit. **(c)** ^{57}Co spectrum reconstructed in energy units.

The centroid channel of each photopeak was mapped to its known energy value, and a linear fit of the form:

$$C = aE + b, \tag{2.9}$$

was applied, where E is the energy in keV and n is the channel number (Fig. 9 (b)). The fit yields $R^2 = 0.999$, confirming a linear channel-to-energy relationship up to at least channel 183, corresponding to an input amplitude of approximately 590 mV. This confirms that the P&H circuit does not introduce amplitude-dependent errors within this bin range.

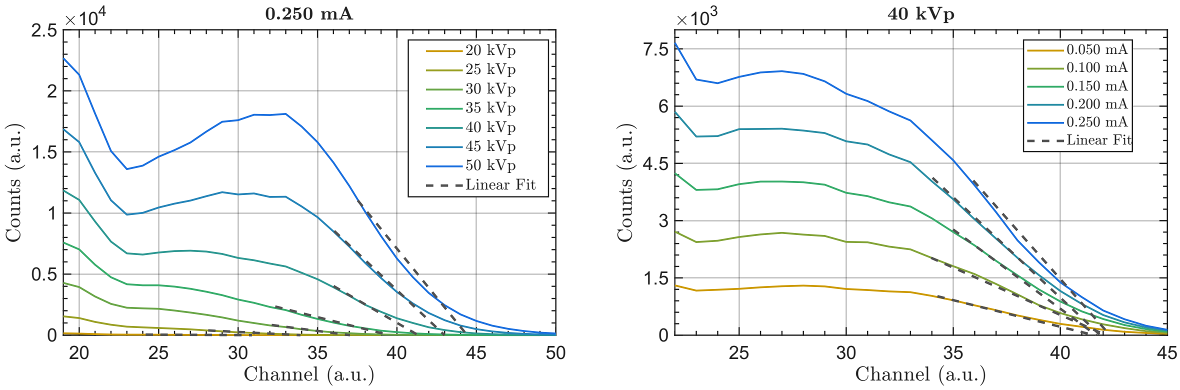
As a validation of the calibration, the ^{57}Co spectrum was reconstructed in energy units and is shown in (Fig. 9 (c)). The fitted centroid of the 122 keV photopeak yields $\mu = 121 \pm 1$ keV, consistent with the nominal value within uncertainty. The measured FWHM is 19 ± 1 keV, corresponding to an energy resolution of $16 \pm 1\%$ at 122 keV, consistent with values reported in the literature for CsI(Tl) [81,82]. This indicates that the front-end electronics do not degrade the intrinsic energy resolution of the crystal. The DORI hardware is therefore considered validated.

Chapter 3. DORI - Calibration

3.1 Energy Calibration

As a novel approach for plastic scintillators, the energy response was calibrated using the *Bremsstrahlung* endpoint of an X-ray tube. The maximum X-ray photon energy produced corresponds to the scenario in which the accelerated electron transfers all its kinetic energy in a single interaction with the anode nucleus. The maximum energy detected, i.e., the endpoint of the acquired spectrum, is therefore equal to the tube voltage, which serves as the energy reference for calibration. Since the *Bremsstrahlung* spectrum near the endpoint is approximately linear, this abscissa can be determined as the x-intercept of a least-squares fit to that region [83].

The detector was positioned outside the beam line, at a distance sufficient to avoid saturation of the front-end electronics. *Bremsstrahlung* spectra were acquired over 5-minute intervals for tube voltages of 15–50 kVp and tube currents of 0.050–0.250 mA, and are shown in Fig. 10 (a) and (b), respectively. Background was subtracted, and channels 18 and below were excluded, as their elevated count rate would otherwise dominate the vertical scale. As represented in Fig. 10 (a), the 15 kVp spectrum is not visible, as it sits within the excluded background. At this distance and fluence, photons of such low energy are absorbed by the black adhesive tape used for light-wrapping of the photosensitive detector element. This establishes the lower detection limit of the front-end electronics at 20 keV.



(a) **(b)**

Figure 10: *Bremsstrahlung* spectra, background subtracted and excluded; **(a)** acquired at 0.250 mA for tube voltages from 15 to 50 kVp; **(b)** acquired at 40 kVp for tube currents from 0.050 to 0.250 mA.

The linear region of each *Bremsstrahlung* spectrum was selected interactively, and a least-squares regression was applied to the points within that window. The x-intercept of the fitted line gives the endpoint channel. Since tube current affects only photon fluence and not the spectral endpoint, the five acquisitions per voltage produce five independent estimates (see Fig. 10 (b)). The resulting values are summarized in Table 7.

Table 7. *Bremsstrahlung* endpoint channel for each tube voltage. Values are the mean of five acquisitions with the maximum deviation from the mean.

Tube Voltage (kVp)	20	25	30	35	40	45	50
Endpoint Channel $\times 10^{-1}$ **(a.u.)**	310 ± 10	334 ± 9	366 ± 4	393 ± 3	419 ± 4	433 ± 5	451 ± 8

Although 20 and 25 keV lie in the boundary at which photoeletric absorption ceases to dominate in BC408 [84], the associated uncertainty is dominated by counting statistics. The associated error improves between 30 and 45 keV, indicating that in this region both the photoelectric cross-section and the photon fluence are sufficient for the endpoint to be determined with good precision. At 50 keV, the photoelectric absorption in the scintillator drops below 0.5%, and consequently the uncertainty increases. As full energy transfer becomes less probable, the spectrum compresses toward a lower and less reliable endpoint. Additionally, pileup further distorts the spectral shape near the abscissa. The pileup distortion is visible as a non-linear tail in the higher energy spectra of Fig. 10 (a), which is not characteristic of filtered *Bremsstrahlung*. At higher count rates, the baseline does not recover between successive pulses, causing them to superimpose and producing the observed tail. These are the fundamental limitations of the endpoint method at higher energies, as the associated uncertainty is expected to grow accordingly.

To extend the calibration beyond the range of the X-ray tube, spectra of the reference sources were acquired with the plastic scintillator five times, each over a 90 min interval. Of the three sources, the only features available are the ^{241}Am photopeak at 59.5 keV, and the Compton edge of the 122 keV line of ^{57}Co, at 39.46 keV, shown in Fig. 11 (a) and (b) respectively. The latter is treated as a validation of the *Bremsstrahlung* method, as it falls within the energy range provided by the tube. The photopeak is, therefore, included as a calibration point at that energy.

A weighted least-squares fit was applied to account for the differences in uncertainty across the energy range and is shown in Fig. 11 (c). Although the 45 keV uncertainty is small, both the 45 and 50 keV endpoints fall below the calibration line, beyond their uncertainties. This underestimation is consistent with the compression of the spectrum discussed above. In contrast, the ^{241}Am centroid falls on the calibration line with an uncertainty smaller than that of the highest-energy endpoint estimate. Although the peak is not as sharp as that obtained with the CsI crystal in Section 2.4, the centroid position is reproducible across acquisitions.

The weighted fit ensures that these better-determined points contribute more to the calibration, and the uncertainties in each estimate propagate into the calibration coefficients [83]. The fit is of the form:

$$C = aE + b, \tag{3.1}$$

where C is the channel number and E is the energy in keV. With $R^2 = 0.989$, the DORI response can be treated as locally linear within the calibrated interval of 20 to 60 keV. Given a measured channel, the energy is therefore obtained as [85]:

$$E = \frac{C - 21}{0.51}, \tag{3.2}$$

$$\sigma_E = \frac{1}{0.51}\sqrt{(0.02)^2\, E^2 + 1^2}. \tag{3.3}$$

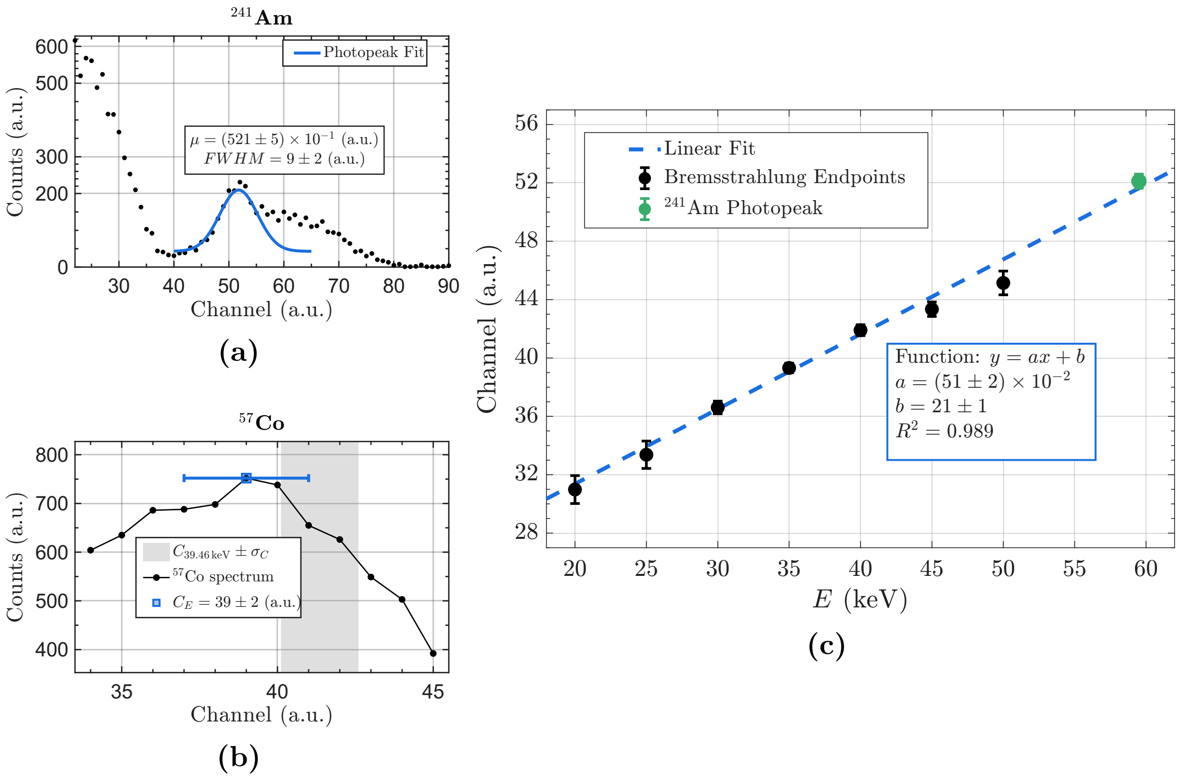


Figure 11: (a) ^{241}Am spectrum, background subtracted, with the Gaussian fit to the 59.5 keV photopeak. The annotated centroid and FWHM are the mean of five acquisitions, with the maximum deviation from the mean as uncertainty. **(b)** ^{57}Co spectrum in the region of the Compton edge of the 122 keV line. The band marks the channel at which the calibration predicts the 39.46 keV edge, within its propagated uncertainty. **(c)** Weighted least-squares calibration fit to the seven *Bremsstrahlung* endpoints and the ^{241}Am photopeak centroid.

As the only available photopeak, the ^{241}Am centroid is used to calculate the energy resolution of a plastic scintillator system at 59.5 keV [79, 80]. Although R is dimensionless and, in principle, independent of the channel-to-energy conversion, the non-zero intercept b shifts the centroid without contributing to the FWHM. Consequently, the resolution differs between channel and energy space and these parameters must be converted to energy using Eqs. 3.2 and 3.3. With the values in Fig. 11 (a), DORI has an energy resolution of $R = (29 \pm 5)\%$ at 59.5 keV, consistent with values reported in literature for BC408 plastic scintillators [68, 80].

As for the Compton edge of the 122 keV line of ^{57}Co, the feature was extracted as the channel of maximum counts. Its uncertainty is larger than that of the ^{241}Am photopeak, due to the less defined nature of the feature. Within these limitations presented in Section 2.1, and as represented in Fig. 11 (b), the extracted channel, C_E, falls, within its own uncertainty, inside the band at which the calibration predicts the 39.46 keV edge. The calibration therefore correctly locates an independent energy reference. The edge was nevertheless excluded from the calibration itself, since the 40 kVp *Bremsstrahlung* endpoint provides a more reliable point at a close energy.

Interpolation within 20 to 60 keV is therefore justified, whereas extrapolation beyond it is not. Extension of the linear fit to 100 keV cannot be assumed, as it would span the non-proportional region of the plastic response discussed in Section 2.1. In the absence of additional calibration points, the gap between 60 and 100 keV is acknowledged as a limitation of the present calibration. Within these limits, the DORI prototype is considered validated in energy across the 20 to 60 keV range. The following dose calibration is performed within this interval.

3.2 Dose Rate Calibration

Chapter 4. DORI - Final Prototype

4.1 Future Work

Chapter 5. Conclusions

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